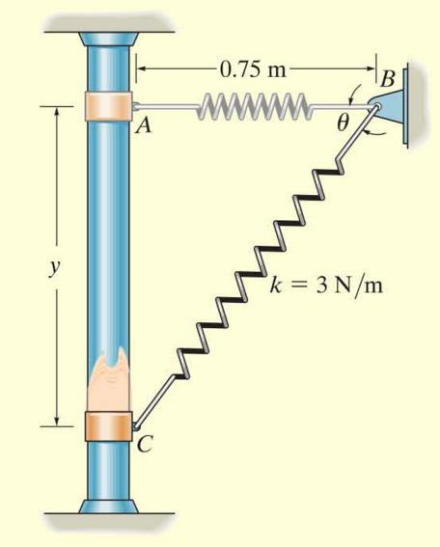
TUTORIAL 3: KINEMATICS OF A PARTICLE

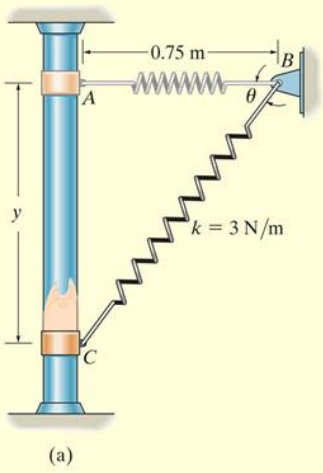
ME 1060: DYNAMICS

Question 1

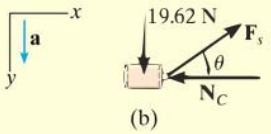
A smooth 2-kg collar C, shown in Fig, is attached to a spring having a stiffness k = 3 N/m and an unstretched length of 0.75 m. If the collar is released from rest at A, determine its acceleration and the normal force of the rod on the collar at the instant y = 1 m.



A smooth 2-kg collar C, shown in Fig. 13–9*a*, is attached to a spring having a stiffness $k = 3$ N/m and an unstretched length of 0.75 m. If the collar is released from rest at A, determine its acceleration and the normal force of the rod on the collar at the instant $y = 1$ m.



(a)



(b)

Fig. 13–9

SOLUTION

Free-Body Diagram. The free-body diagram of the collar when it is located at the arbitrary position y is shown in Fig. 13–9*b*. Furthermore, the collar is *assumed* to be accelerating so that "**a**" acts downward in the *positive* y direction. There are four unknowns, namely, N_C, F_s, a, and θ.

Equations of Motion.

$$\stackrel{+}{\rightarrow} \Sigma F_x = ma_x; \qquad -N_C + F_s \cos\theta = 0 \tag{1}$$

$$+\downarrow \Sigma F_y = ma_y; \qquad 19.62 - F_s \sin\theta = 2a \tag{2}$$

From Eq. 2 it is seen that the acceleration depends on the magnitude and direction of the spring force. Solution for N_C and a is possible once F_s and θ are known.

The magnitude of the spring force is a function of the stretch s of the spring; i.e., $F_s = ks$. Here the unstretched length is $AB = 0.75$ m, Fig. 13–9*a*; therefore, $s = CB - AB = \sqrt{y^2 + (0.75)^2} - 0.75$. Since $k = 3$ N/m, then

$$F_s = ks = 3\left(\sqrt{y^2 + (0.75)^2} - 0.75\right) \tag{3}$$

From Fig. 13–9*a*, the angle θ is related to y by trigonometry.

$$\tan\theta = \frac{y}{0.75} \tag{4}$$

Substituting $y = 1$ m into Eqs. 3 and 4 yields $F_s = 1.50$ N and $\theta = 53.1°$. Substituting these results into Eqs. 1 and 2, we obtain

$$N_C = 0.900 \text{ N} \qquad \textit{Ans.}$$

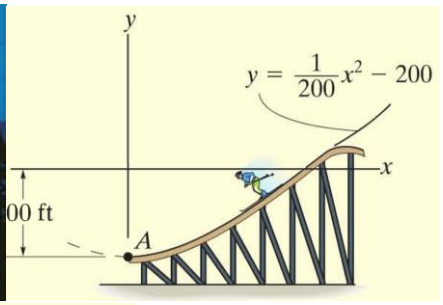
$$a = 9.21 \text{ m/s}^2 \downarrow \qquad \textit{Ans.}$$

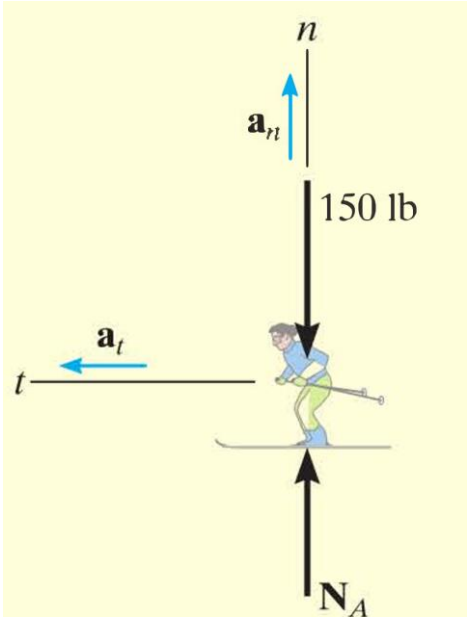
NOTE: This is not a case of constant acceleration, since the spring force changes both its magnitude and direction as the collar moves downward.

Question 2

Design of the ski jump shown in the photo requires knowing the type of forces that will be exerted on the skier and her approximate trajectory. If in this case the jump can be approximated by the parabola shown in figure, determine the normal force on the $150 - lb$ skier the instant she arrives at the end of the jump, point A, where her velocity is $65\ ft/s$. Also, what is her acceleration at this point?







Free-Body Diagram. Since $dy/dx = x/100\,|_{x=0} = 0$, the slope at A is horizontal. The free-body diagram of the skier when she is at A is shown in Fig. 13–14*b*. Since the path is *curved*, there are two components of acceleration, $\mathbf{a}_n$ and $\mathbf{a}_t$. Since a_n can be calculated, the unknowns are a_t and N_A.

Equations of Motion.

$$+\uparrow \Sigma F_n = ma_n; \qquad N_A - 150 = \frac{150}{32.2}\left(\frac{(65)^2}{\rho}\right) \tag{1}$$

$$\overset{+}{\leftarrow} \Sigma F_t = ma_t; \qquad 0 = \frac{150}{32.2}a_t \tag{2}$$

The radius of curvature ρ for the path must be determined at point $A(0, -200\text{ ft})$. Here $y = \frac{1}{200}x^2 - 200$, $dy/dx = \frac{1}{100}x$, $d^2y/dx^2 = \frac{1}{100}$, so that at $x = 0$,

$$\rho = \left.\frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}\right|_{x=0} = \frac{[1 + (0)^2]^{3/2}}{|\frac{1}{100}|} = 100\text{ ft}$$

Substituting this into Eq. 1 and solving for N_A, we obtain

$$N_A = 347\text{ lb} \qquad \textit{Ans.}$$

Kinematics. From Eq. 2,

$$a_t = 0$$

Thus,

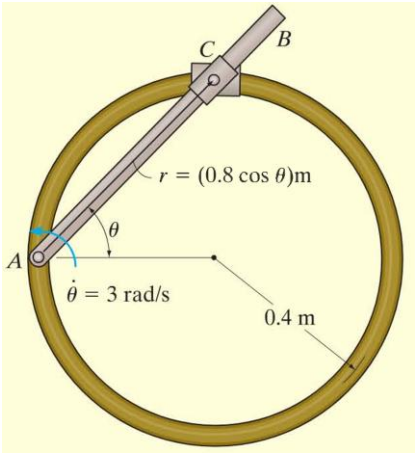
$$a_n = \frac{v^2}{\rho} = \frac{(65)^2}{100} = 42.2\text{ ft/s}^2$$

$$a_A = a_n = 42.2\text{ ft/s}^2\uparrow \qquad \textit{Ans.}$$

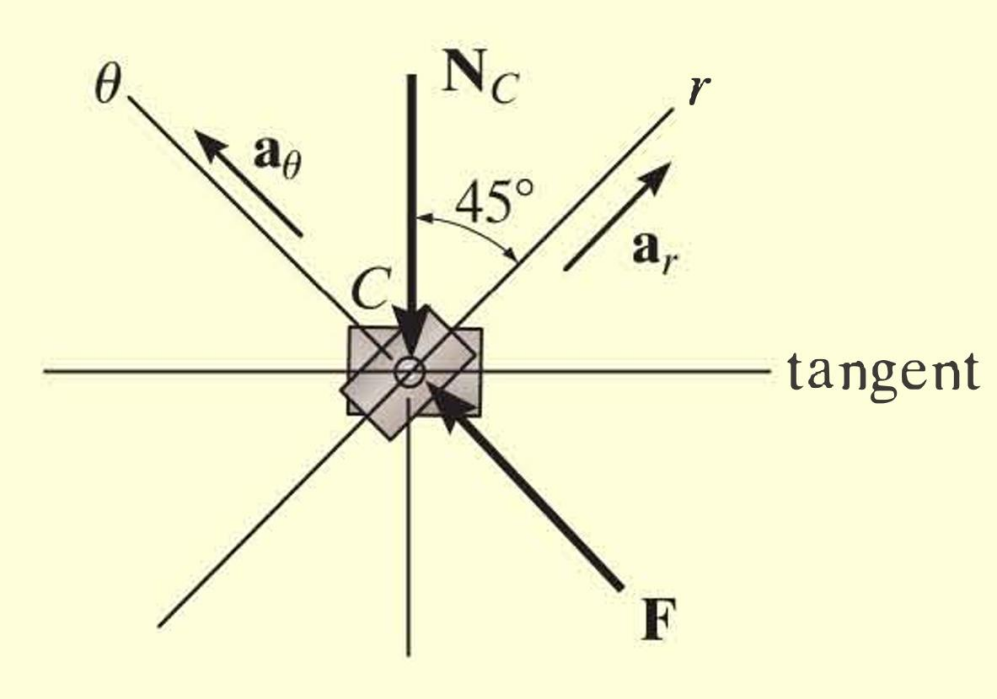
NOTE: Apply the equation of motion in the y direction and show that when the skier is in midair her acceleration is 32.2 ft/s^2.

Question 3

The smooth $0.5 - kg$ double-collar in figure can freely slide on arm AB and the circular guide rod. If the arm rotates with a constant angular velocity of $\dot{\theta}$ = 3 rad/s, determine the force the arm exerts on the collar at the instant θ = 45°. Motion is in the horizontal plane.



The normal reaction Nc of the circular guide rod and the force F of arm AB act on the collar in the plane of motion. Note that F acts perpendicular to the axis of arm AB, that is, in the direction of the θ axis, while Nc acts perpendicular to the tangent of the circular path at θ = 45°. The four unknowns are $Nc,\ F,\ a_n\ a_\theta$



Equation of Motion:

$$\sum F_r = ma_r; \quad -N_C \cos 45 = 0.5\, a_r$$

$$\sum F_\theta = ma_\theta; \quad F - N_C \sin 45 = 0.5\, a_\theta$$

Kinematics

$$r = 0.8 \cos\theta = 0.8 \cos 45 = 0.5657\ m$$

$$\dot{r} = -0.8 \sin\theta\ \dot{\theta} = -0.8 \sin 45\ (3) = -1.6971\ ms^{-1}$$

$$\ddot{r} = -0.8\left[\sin\theta\ \ddot{\theta} + \cos\theta\ \dot{\theta}^2\right] = -0.8\left[\sin 45\ (0) + \cos 45\ (3)^2\right] = -5.091\ ms^{-2}$$

We have;

$$a_r = \ddot{r} - r\dot{\theta}^2 = \text{-5.091 m/s\^2 - (0.5657 m)(3 rad/s)2 = -10.18 m/s2}$$

$$a_\theta = r\,\ddot{\theta} + 2\dot{r}\dot{\theta} = (0.5657)(0) + 2(-1.697)(3) = -10.18\ ms^{-2}$$

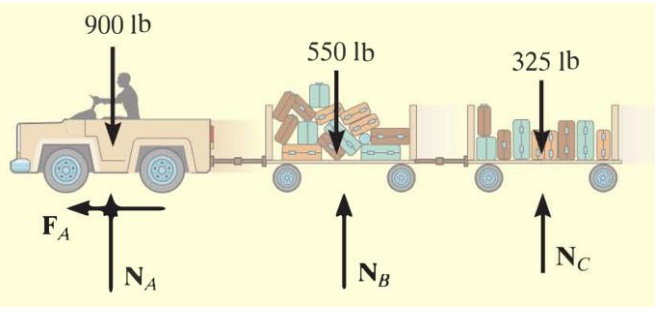
Substituting these results, we get,

$N_C = 7.2\ N$ and $F = 0$

Question 4

The baggage truck A shown in the photo has a weight of $900\ lb$ and tows a $550-lb$ cart B and a $325-lb$ cart C. For a short time, the driving frictional force developed at the wheels of the truck is $F_A = (40t)\ lb$, where t is in seconds. If the truck starts from rest, determine its speed in 2 seconds. Also, what is the horizontal force acting on the coupling c between the truck and cart B at this instant? Neglect the size of the truck and carts.





(a)

SOLUTION

Free-Body Diagram. As shown in Fig. 13–8*a*, it is the frictional driving force that gives both the truck and carts an acceleration. Here we have considered all three vehicles as a single system.

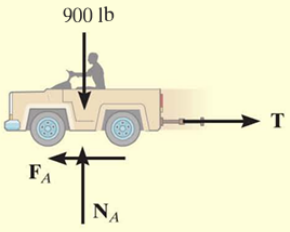
Equation of Motion. Only motion in the horizontal direction has to be considered.

$$\stackrel{+}{\leftarrow} \Sigma F_x = ma_x; \qquad 40t = \left(\frac{900 + 550 + 325}{32.2}\right)a$$

$$a = 0.7256t$$

Kinematics. Since the acceleration is a function of time, the velocity of the truck is obtained using $a = dv/dt$ with the initial condition that $v_0 = 0$ at $t = 0$. We have

$$\int_0^v dv = \int_0^{2\,s} 0.7256t\, dt; \quad v = 0.3628t^2\Big|_0^{2\,s} = 1.45\ \text{ft/s} \qquad \textit{Ans.}$$



(b)

Fig. 13–8

Free-Body Diagram. In order to determine the force between the truck and cart B, we will consider a free-body diagram of the truck so that we can "expose" the coupling force **T** as external to the free-body diagram, Fig. 13–8*b*.

Equation of Motion. When $t = 2$ s, then

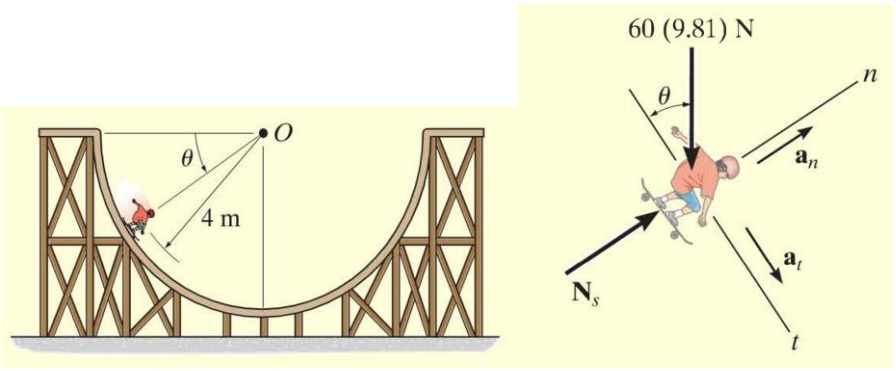
$$\stackrel{+}{\leftarrow} \Sigma F_x = ma_x; \qquad 40(2) - T = \left(\frac{900}{32.2}\right)[0.7256(2)]$$

$$T = 39.4\ \text{lb} \qquad \textit{Ans.}$$

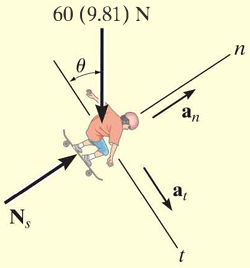
NOTE: Try and obtain this same result by considering a free-body diagram of carts B and C as a single system.

Question 5

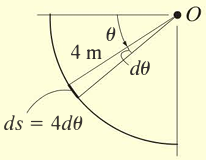
The 60-kg skateboarder in Fig.13-15a coasts down the circular track. If he starts from rest when e = 0°, determine the magnitude of the normal reaction the track exerts on him when e = 60°. Neglect his size for the calculation.



(a)



(b)



(c)

Fig. 13–15

SOLUTION

Free-Body Diagram. The free-body diagram of the skateboarder when he is at an *arbitrary position* θ is shown in Fig. 13–15*b*. At $\theta = 60°$ there are three unknowns, N_s, a_t, and a_n (or v).

Equations of Motion.

$$\downarrow \Sigma F_n = ma_n; \quad N_S - [60(9.81)\text{N}] \sin\theta = (60\text{ kg})\left(\frac{v^2}{4\text{m}}\right) \tag{1}$$

$$\downarrow \Sigma F_t = ma_t; \quad [60(9.81)\text{N}] \cos\theta = (60\text{ kg})\, a_t$$

$$a_t = 9.81 \cos\theta$$

Kinematics. Since a_t is expressed in terms of θ, the equation $v\,dv = a_t\,ds$ must be used to determine the speed of the skateboarder when $\theta = 60°$. Using the geometric relation $s = \theta r$, where $ds = r\,d\theta = (4\text{ m})\,d\theta$, Fig. 13–15*c*, and the initial condition $v = 0$ at $\theta = 0°$, we have,

$$v\,dv = a_t\,ds$$

$$\int_0^v v\,dv = \int_0^{60°} 9.81 \cos\theta (4\,d\theta)$$

$$\frac{v^2}{2}\bigg|_0^v = 39.24 \sin\theta \bigg|_0^{60°}$$

$$\frac{v^2}{2} - 0 = 39.24(\sin 60° - 0)$$

$$v^2 = 67.97\text{ m}^2/\text{s}^2$$

Substituting this result and $\theta = 60°$ into Eq. (1), yields

$$N_s = 1529.23\text{ N} = 1.53\text{ kN} \qquad \textit{Ans.}$$